

Verdant Coil Git Usage Strategy & Workflow

Purpose

This document outlines the standard operating procedure (SOP) for using GitHub effectively during the development of *Verdant Coil*. It ensures consistent version control, safe backups, and smooth collaboration between devices and contributors.

Core Objectives

- Maintain **safe, sequential backups** of all progress.
 - Encourage **modular, testable development** using feature branches.
 - Enable **easy rollback and debugging**.
 - Provide clear **historical records** of changes and milestone completions.
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General Git Workflow

1. `main` Branch = Stable Builds Only

- `main` (or `master`) must always reflect a functional, crash-free version.
- No direct commits to `main` unless it is a tested hotfix.

2. Create a New Branch for Each Feature or Fix

- Always create a branch for each new task:
- `feature/level-builder`
- `bugfix/hazard-damage`
- `experiment/turn-order`
- Purpose: isolate work and prevent breaking stable code.

3. Commit Often with Meaningful Messages

- Commit after each logical change.
- Messages should summarize *what* and *why*:
- `Add DigestWall interaction logic`
- `Fix crawler skipping hazard on move`

4. Merge to `main` When Complete

- Only merge after testing and confirming the feature works.
- Use squash merge for a clean history:

```
git checkout main
git merge --squash feature/level-builder
```

5. Tag Milestones

- Use tags to mark completed phases:

```
git tag phase-1.1-mvp
```

- Enables quick rollback or reference.

Debugging Strategy

- Use throwaway branches for risky tests:
- `experiment/input-handler`
- If successful, merge into the real feature branch.
- If not, delete the branch with no harm done.

Naming Conventions

Task Type	Branch Format
Feature	<code>feature/<description></code>
Bug Fix	<code>bugfix/<description></code>
Performance	<code>perf/<description></code>
UI	<code>ui/<description></code>
Temporary/Test	<code>experiment/<description></code>

Multi-Device Usage

If switching devices:

1. **Push** current branch at end of session.
2. On new device: **pull and checkout** correct branch.
3. If branches diverged, backup with:
4. `old-laptop-feature-x`

Safety Rules

- Do **not** force push unless absolutely required.
 - Only delete branches once fully merged.
 - Periodically backup the GitHub repo externally.
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Optional: Documentation

Maintain a `CHANGELOG.md` or use commit messages to document:

- Active development phase (e.g. Phase 1.1)
 - New features and fixes
 - Known issues
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This SOP ensures orderly, recoverable development progress for *Verdant Coil* and supports the collaborative, exploratory nature of the project.